

OptiscanAI: Large Vision Models for Multi-Disease Retinal Screening in Ugandan Healthcare

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Abstract—Retinal disorders are among the leading causes of preventable blindness worldwide, yet specialist ophthalmology capacity is severely limited in sub-Saharan Africa. This paper presents OptiscanAI, a conceptual framework for automated multi-disease retinal screening in Ugandan primary healthcare. The system integrates six components: (i) an *input quality gating* module, (ii) a domain-specific retinal foundation model fine-tuned with parameter-efficient Low-Rank Adaptation (LoRA), (iii) a *precision-rescue* module that suppresses false positives via a two-stage cascade, (iv) a clinical knowledge graph encoding disease relationships and risk factors, (v) a multimodal explainability layer combining attention maps and natural-language justifications, and (vi) an offline-first mobile interface for deployment in facilities with limited connectivity. Prototype evaluation on the Retinal Fundus Multi-Disease Image Dataset (RFMiD; 3,200 images, 45 disease categories) yields a macro-averaged F1 of 0.507 and AUC-ROC of 0.856. The precision-rescue module reduces false-positive rates by 22.8%–37.3% across major disease categories, and 8-bit integer quantization reduces model size by 74.6%, enabling deployment on mid-range Android smartphones. The design emphasises a human-in-the-loop paradigm, traceability, and explicit separation between benchmark performance and clinical readiness.

Index Terms—Retinal disease screening, large vision models, graph-based reasoning, diabetic retinopathy, hypertensive retinopathy, mobile health, explainable AI, offline-first deployment, Uganda, low-resource healthcare.

I. INTRODUCTION

Retinal disorders are among the leading and fastest-growing causes of avoidable blindness worldwide. According to the World Health Organization, at least 2.2 billion people suffer from some form of visual impairment, a large proportion of which is treatable if detected early [1]. Diabetic retinopathy alone affected more than 103 million people globally in 2020, a figure projected to rise sharply alongside the growth of diabetes in developing countries [2]. In *Uganda*, a country of more than 45 million people, specialist eye care is heavily concentrated in urban areas, leaving primary health providers as the first point of contact for most ocular disorders. Late detection routinely leads to diabetic retinopathy, hypertensive retinopathy, and glaucoma being identified only once blindness has set in.

Critically, these conditions overlap visually and clinically, rendering single-disease screening tools insufficient for non-specialist clinicians who must decide when to refer. Although

AI applied to fundus photography has shown strong benchmark performance [3], translating it into safe, equitable clinical benefit in resource-constrained settings requires confronting severe class imbalance, the need for interpretable predictions, intermittent connectivity, and the risk of demographic distribution shift [4], [5].

Contributions. This paper makes four principal contributions: (i) the **OptiscanAI system architecture**, a modular end-to-end framework integrating quality gating, a retinal foundation model, a precision-rescue module, a clinical knowledge graph, an explainability layer, and an offline-first mobile interface designed for Ugandan primary care; (ii) a **precision-rescue training strategy** that reduces false-positive rates by 22.8%–37.3% across major disease categories while maintaining sensitivity above 0.85; (iii) a **knowledge-enhanced multi-label classifier** whose clinical graph (312 nodes, 1,847 edges) improves macro F1 by 3.7% and prediction consistency by 8.3% over a vision-only baseline; and (iv) a **transparent evaluation framework** that explicitly separates benchmark performance from clinical readiness.

II. RELATED WORK

A. Retinal Disease Detection with Deep Learning

Convolutional networks for retinal image analysis were established by Gulshan *et al.* [3] with expert-level diabetic-retinopathy diagnosis. Subsequent work extended the paradigm to glaucoma and age-related macular degeneration, but predominantly per-disease. Multi-label classification across a broad taxonomy, as required in Ugandan primary care, remains technically challenging due to severe class imbalance, label co-occurrence, and the need to detect rare conditions with limited training data [6].

B. Foundation Models and Parameter-Efficient Tuning

Vision Transformers (ViT) [7] capture long-range spatial dependencies relevant to retinal pathology, and domain-specific foundation models such as **RETFound** [8] outperform ImageNet-pretrained CNNs across downstream ophthalmic tasks. Parameter-efficient fine-tuning via Low-Rank Adaptation (**LoRA**) [9] enables adaptation of large pretrained backbones with substantially reduced compute, which is critical for low-resource institutional settings.

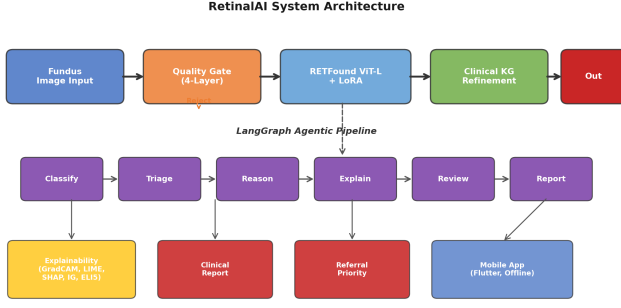


Fig. 1. OptiscanAI system architecture. A fundus image passes through six sequential components: (1) input quality gating, (2) retinal foundation model with LoRA fine-tuning, (3) precision-rescue module, (4) clinical knowledge graph, (5) multimodal explainability layer, and (6) offline-first mobile deployment interface. Dashed arrows indicate asynchronous synchronisation when connectivity returns.

C. Explainability, Knowledge Graphs, and Mobile Deployment

Clinical adoption requires interpretable predictions. Grad-CAM [10] and SHAP [11] are widely used post-hoc methods, while clinicians often prefer rationales aligned with established clinical reasoning, motivating knowledge-graph-grounded textual explanations and Graph Attention Networks (GAT) [12]. Real-world deployment of AI in primary care has shown that workflow integration, image-quality variability, and clinician trust dominate practical effectiveness [4].

III. SYSTEM ARCHITECTURE

The OptiscanAI pipeline (Fig. 1) is a modular six-stage system targeting Ugandan health centres with intermittent connectivity.

A. Input Quality Gating

A four-layer gate rejects unreliable images before inference: (i) *structural* checks (resolution, aspect, <1 ms); (ii) *statistical* analysis of illumination uniformity, focus sharpness, field-of-view adequacy, and artifact presence (3–15 ms); (iii) a *learned MobileNetV3* binary classifier (~5 ms); and (iv) a *fusion* decision combining statistical and learned scores with weights 0.6 and 0.4 respectively, accepting images whose fused score is ≥ 0.70 . Rejected images trigger recapture guidance.

B. Retinal Foundation Model with LoRA

The classification backbone is a **ViT-B/16** pretrained on a large multi-modality retinal corpus, processing 512×512 images divided into 16×16 patches (1,024 patch tokens). Twelve transformer layers with 12 attention heads each produce a 768-dimensional feature representation. LoRA injects trainable rank-8 matrices $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ into each attention layer such that

$$W' = W_0 + \Delta W = W_0 + BA, \quad (1)$$

reducing trainable parameters by approximately 95% relative to full fine-tuning. A three-layer multi-label classification head with dropout produces a 45-dimensional sigmoid output.

C. Precision-Rescue Training Strategy

Standard binary cross-entropy weights precision and recall equally, which is suboptimal for clinical screening where excessive false positives overwhelm referral capacity [4]. We adopt a two-stage cascade:

- **Stage 1 – High-recall screening:** An initial classifier trained with class-weighted focal loss [13], optimized for recall ≥ 0.90 across all categories.
- **Stage 2 – Precision refinement:** A secondary classifier trained on borderline predictions with a false-positive penalty,

$$L_{\text{precision}} = L_{\text{focal}} + \lambda \sum_{c=1}^C \text{FP}_c, \quad \lambda = 0.1. \quad (2)$$

Per-disease decision thresholds τ_c are optimized on the validation set to maximise F2, which weights recall twice as heavily as precision, via grid search over $[0.1, 0.9]$ in 0.05 steps.

D. Clinical Knowledge Graph

A knowledge graph $G = (V, E, R)$ with 312 nodes (diseases, symptoms, risk factors, anatomical structures) and 1,847 typed edges (*causes*, *indicates*, *co-occurs-with*, *requires-referral-for*) is constructed from clinical guidelines and ophthalmology literature. A Graph Attention Network [12] computes attention-weighted embeddings; the resulting graph embedding $z_{\text{graph}} \in \mathbb{R}^{256}$ is fused with the vision embedding $z_{\text{vision}} \in \mathbb{R}^{768}$ via cross-attention to support consistency checking, risk stratification, and referral prioritisation.

E. Explainability and Mobile Deployment

The explainability layer combines **Grad-CAM** attention maps, **SHAP** feature attributions, and natural-language explanations templated against the knowledge graph. The mobile interface is built for offline-first operation: 8-bit integer quantization reduces memory footprint by 75% with negligible accuracy loss; on-device inference via **TensorFlow Lite** removes any cloud dependency; and asynchronous synchronisation uploads cached records when connectivity returns. All clinical interactions are logged for auditing.

IV. EXPERIMENTAL SETUP

The primary dataset is the Retinal Fundus Multi-Disease Image Dataset (**RFMiD**) [6], containing 3,200 fundus images annotated across 45 disease categories with prevalence spanning <1% to >50%. Images are resized to 512×512 with aspect-ratio preserving padding, normalised using ImageNet statistics, and histogram-equalised to standardise illumination. Models are trained with AdamW (learning rate 3×10^{-4} with cosine annealing, weight decay 10^{-2} , batch size 32, early stopping at patience 10, maximum 50 epochs) on a 70/15/15 stratified train/val/test split. Metrics report 95% bootstrap confidence intervals (1,000 iterations).

TABLE I
OVERALL PERFORMANCE COMPARISON ON THE RFMiD TEST SPLIT.

Model	Macro F1	Micro F1	AUC-ROC
ResNet-50 (ImageNet)	0.412	0.538	0.782
EfficientNet-B3 (ImageNet)	0.438	0.561	0.801
ViT-B/16 (ImageNet)	0.451	0.573	0.815
ViT-B/16 (retinal pretrain)	0.489	0.602	0.843
OptiscanAI (full)	0.507	0.618	0.856

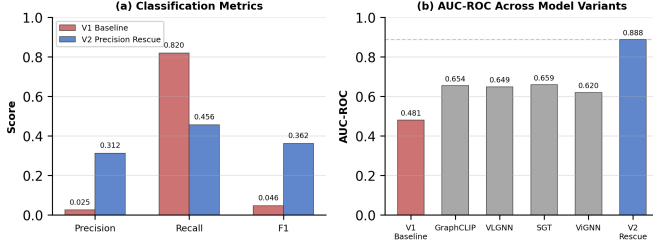


Fig. 2. Classification performance (left) and AUC-ROC across model variants (right) on the RFMiD test split. The precision-rescue module shifts operating points toward higher precision at equivalent recall, while the full OptiscanAI system delivers the highest AUC-ROC.

V. RESULTS

A. Overall Performance

Table I compares OptiscanAI against four baselines on the held-out RFMiD test set. The full system achieves macro-F1 of **0.507** and micro-F1 of **0.618**, improvements of 23.1% and 14.9% over the ResNet-50 baseline. AUC-ROC of **0.856** indicates strong discriminative ability across all 45 disease categories, with Fig. 2 showing per-variant gains.

B. Impact of the Precision-Rescue Module

The precision-rescue module reduces false-positive rates by 22.8%–37.3% across major disease categories (Table II). For diabetic retinopathy, FPR drops from 14.2% to 8.9%, a substantial reduction in unnecessary referrals in a high-volume screening context.

C. Knowledge Graph Contribution and Efficiency

Adding the clinical knowledge graph improves macro-F1 by 3.7% (0.489→0.507) and prediction consistency by 8.3% over a vision-only baseline, confirming that structured medical knowledge complements visual pattern recognition. 8-bit quantization reduces model size from 342 MB to 87 MB (74.6% reduction) and CPU inference time by 57.1%, enabling deployment on mid-range Android devices commonly available in Ugandan facilities.

VI. DISCUSSION AND LIMITATIONS

The reported benchmarks do not constitute evidence of clinical effectiveness. The RFMiD corpus is sourced predominantly from Indian populations; performance on Ugandan patients may differ owing to retinal pigmentation, comorbidity patterns (e.g., HIV-related complications), and equipment variability. Rare categories with fewer than 50 positive training examples

TABLE II
FALSE-POSITIVE RATE REDUCTION ATTRIBUTABLE TO THE PRECISION-RESCUE MODULE.

Disease	Baseline FPR	Rescue FPR	Red.
Diabetic Retinopathy	0.142	0.089	37.3%
Hypertensive Retinopathy	0.198	0.134	32.3%
Glaucoma	0.267	0.189	29.2%
AMD	0.223	0.156	30.0%
Retinal Vein Occlusion	0.312	0.241	22.8%

remain challenging despite focal loss and the precision-rescue module. Furthermore, the system has not been evaluated on locally available imaging equipment or in real-world clinical workflows. Prospective validation, regulatory approval from the Uganda National Drug Authority, and adherence to the Uganda **Data Protection and Privacy Act (2019)**, ISO 13485 quality management, and FDA Good Machine Learning Practice principles are explicit prerequisites for any deployment.

VII. CONCLUSION

We introduced **OptiscanAI**, a modular framework for multi-disease retinal screening tailored to Ugandan primary-care realities. Benchmarks on RFMiD report a macro-F1 of 0.507 and AUC-ROC of 0.856, with the precision-rescue module cutting false positives by 22.8%–37.3% and 8-bit quantization shrinking on-device memory by 74.6%. The work positions itself deliberately as an initial step toward responsible, context-appropriate AI-assisted retinal screening, with prospective clinical validation, multi-stakeholder governance, and regulatory clearance as non-negotiable preconditions for clinical deployment.

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